Technical Analysis Report
Group 10

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Introduction

This report aims to show and explain the detailed analysis process for modelling the system of the solar panel array design, where certain outputs were extracted to help inform the whole design process. This report will include the assumptions made, estimations and clearly defined worst cases to incorporate safety considerations.

Initial Model for Deployment Phase

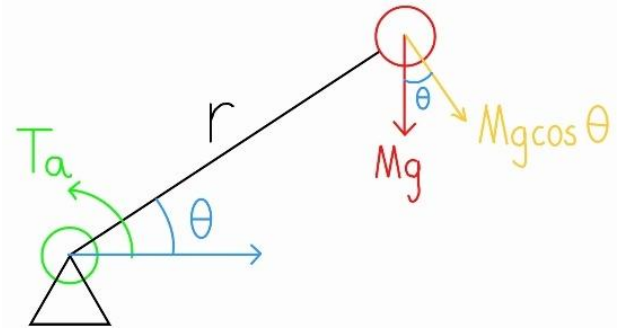


Figure 1- Simple Model of Inverted Pendulum

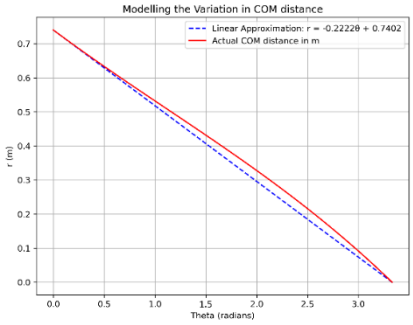


Figure 2 - COM distance variation and Approximation

The system's behavior was assumed to be that of an inverted pendulum (figure 1), where a centre of mass is at a distance from the system's main pivot point. This approximation could produce a model which portrays the linkage system's dynamics relatively accurately. The following equation of motion [1] was used to approximate the system:

$$\Sigma T_a - \Sigma T_r = I\ddot{\theta} \quad [1]$$

Where $T_a$is the accelerating torque in Nm, T_r is the resistive torque, I is the moment of Inertia in $kg/m^2$and $\ddot{\theta}$ is the angular acceleration in of the pendulum in rad/s^2

T_a will be the torque generated by the motor, and T_r will be the torque due to the component weight of the centre of mass perpendicular to r. Damping was considered in the retraction phase only due to the assumption that the deployment phase will not need damping to reduce its speed if an optimal gear ratio was chosen. Instead, it was opted to assume there will be no resistive torque due to friction from the joints as the design includes bearings which will produce minimal resistive force.

T_a is the accelerating torque which was dependent on the motor and gearbox ratio. Thus motor variability [2] had to be considered for each motor in order to accurately model the system's behavior for each motor

$$T_a = \left(\left(-\frac{T_{stall}}{\omega_{no\ load}}\right)(\omega_{load} \times VR) - T_{stall}\right)VR \quad [2]$$

Where T_{stall} was the motor stall torque in Nm, $\omega_{no\ load}$ was the angular velocity of the motor with no load in rad/s, ω_{load} was the angular velocity of the centre of mass through its deployment and VR was the velocity ratio of the whole gearbox. The ω_{motor} was dependent on the velocity ratio itself and the angular velocity of the load the motor is trying rotate so the equation $\omega_{motor} = \omega_{load} \times VR$ was incorporated into the formula

$$T_r = Mgcos\theta \times r \quad [3]$$

Where M was the total mass of the system in kg, g was the gravitational acceleration of 9.81 m/s, θ was the angle of deployment in radians and r is the distance of centre of mass from pivot in m. Total mass was estimated using the density of stainless steel and volume of each individual linkage. One of the assumptions made was that r varied with angle of deployment (figure 2). In order to evaluate this, the centre of mass position had to be analysed thoroughly. the centre of mass position was recorded from the 3D model assembly, measured at the start and end of the deployment phase, which was at 8.3 degrees initially and 81.5 degrees fully deployed. The following formula [4] was used:

$$x_{COM} = \frac{\Sigma m_i x_i}{\Sigma m_i}\ ,\ y_{COM} = \frac{\Sigma m_i y_i}{\Sigma m_i} \quad [4]$$

Where $x.y,z$ are the coordinates and $m_i$was the mass of ith object (linkages and solar panels). r was computed using measured data where the centre of mass position was calculated at different deployed angles and modelled r to be a linear function as it closely matched the real position of its centre of mass. This resulted in the function of r in terms of $\theta, r = -0.2222\theta + 7.402$. The moment of inertia I of the system was assumed constant as from testing with the CAD model, it was approximated that the distance of each linkage from the main pivot did not vary significantly therefore according to the parallel axis theorem, moment of inertia varied minimally. The following equations [5] were used to find the moment of inertia of each linkage and its inertia around the main axis:

$$I_{COM,i} = \frac{1}{12} m_i L_i\ , I_i = I_{COM,i} + m_i {d_i}^2\ , I_{total} = \Sigma I_i \quad [5]$$

Where m_i is the mass of each linkage in m, L_i is the length of each linkage in m and d_i is the distance of each linkage's COM from main pivot. This resulted in a moment of inertia of 60.8 kg/m^2.

Modelling Motor Efficiency

Each motor considered also had their efficiencies modelled to inform the motor selection. The equation for motor efficiency [6]:

$$Motor\ Efficiency = \frac{Output\ Power}{Input\ Power} = \frac{\tau\omega}{VI} \quad [6]$$

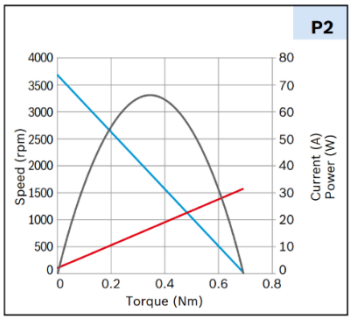


Figure 3 - Performance Curve of NSA-I P2

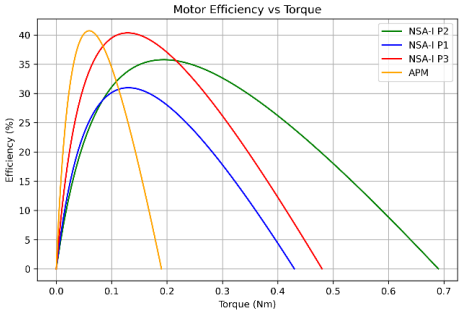


Figure 4 – Motor Efficiencies for selected motors

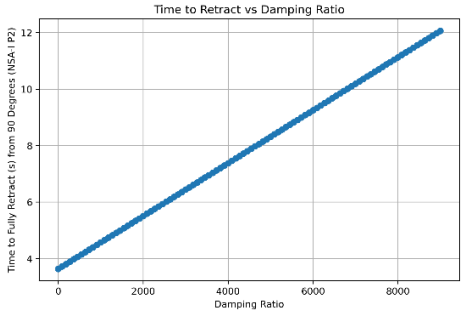


Figure 5 – Time Taken Against Damping Ratios

Where τ is the torque of the motor in Nm, ω is the angular velocity of the motor in rad/s, V is the output voltage of the motor in V (12V for all the motors), and I is the variable current of the motor in A. The performance curves of each motor (APM, NSA-I P1,P2,P3) was given in the Bosch Motor Catalogue (figure 3) which showed how rpm and current varied with torque. Inputting all the variables produced the graphs of motor efficiency for each motor (figure 4).

Model for Retraction Phase (Damping)

For modelling the dynamics of the system retracting, damping was considered in order to match the retraction time to deployment time for the system (11s). When the inverted pendulum is rotating in the other direction, the weight torque component of the system will become an accelerative torque, so the accelerative torque and resistive torque becomes [7]:

$$T_a = \left(\left(-\frac{T_{stall}}{\omega_{no\ load}}\right)(\omega_{load} \times VR) - T_{stall}\right)VR + (Mgcos\theta \times r), \qquad T_r = -c\omega_{load} \quad [7]$$

Where c is the damping coefficient in Nms/rad. The damping coefficient needed for the system was 8000 Nms/rad (inferred from figure 5), where the optimal damper from the ENIDINE catalogue was selected. The reason for using a linear damper as opposed to rotary is because a rotary damper is unable to achieve such high damping coefficients without being excessively expensive. It was decided to just use one damper on one side of the system. This aims to reduce the weight of the entire system and reduce the amount of volume taken up by the damper and crank which could influence the aerodynamics (related to PDS items 13 which minimizes weight and item 7 which ensures aerodynamics). In order to convert rotary movement into linear movement, a crank system must be designed (figure 7). To find the linear damping coefficient needed, the following formula was used [8]:

$$c_{linear} = \frac{c_{rotary}}{r^2} \quad [8]$$

Where c_{linear} is in Ns/m, c_{rotary} is in Nms/rad and r is the moment arm in m. Since a crank design was opted for, the distance where the crank joins onto the main linkage arm to the main pivot was r, and a distance of 0.5m was chosen as to fit the damper onto the design optimally and reduce the damping coefficient needed. Thus c_{linear} was calculated to be 32,000 Ns/m. To find the damping coefficient of a linear damper, equation [9] was used:

$$c_{linear} = \frac{F_{max}}{v_{at\ max}} \quad [9]$$

Where F_{max} is the maximum force the damper can exert in N and v is the velocity of the damper which achieves the maximum force. Most suitable damper was determined to be LD750 as its F_{max} is 7,500N and it reaches that force at around 0.25 m/s. This was optimal as when modelling the retraction phase of the system, the angular velocity peaks at 0.52 rad/s (figure 6) which means a linear velocity of 0.266 m/s at 0.5m from pivot.

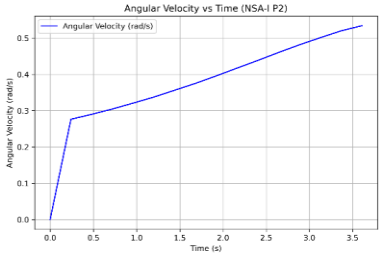


Figure 6 – Angular Velocity (Retracting)



Figure 7 – Damper Placement on Design